

Medical Keyword Extraction Documentation

1. Project Overview

1.1. Goal

The objective of this project is to develop and evaluate a robust pipeline for automatically extracting essential keywords from a large corpus of medical transcriptions. The primary goal is to compare the effectiveness of classical, pre-trained, and fine-tuned models to identify the optimal strategy for this domain-specific Natural Language Processing (NLP) task.

1.2. Dataset

The project utilizes the "Medical Transcriptions" dataset from Kaggle, available at <https://www.kaggle.com/datasets/tboyle10/medicaltranscriptions/data>

2. Dataset Understanding

The dataset is a single CSV file containing transcribed medical reports.

- **Structure:** A flat table with columns representing different attributes of each transcription.
- **Features (Inputs):** The primary feature used is the transcription column, which contains the raw text of the medical report. Other contextual features like `medical_specialty` are available but were not used for model training to ensure a generalized approach.
- **Labels (Outputs):** The keywords column serves as the ground-truth label for our supervised and evaluation tasks. It is a comma-delimited string of human-annotated keywords relevant to the corresponding transcription.

3. Data Preprocessing

A rigorous and standardized preprocessing pipeline was established to ensure data quality, consistency, and compatibility with the models.

Steps:

1. **Loading & Selection:** The `mtsamples.csv` file was loaded into a pandas DataFrame. Only the essential transcription and keywords columns were selected.
2. **Handling Missing Values:** Rows with null values in the selected columns were dropped using `.dropna()` to prevent errors and ensure every record was a complete pair of text and labels.

3. Text Cleaning (clean_text function):

- **Lowercasing:** All transcription text was converted to lowercase to ensure uniformity
- **Whitespace Normalization:** All sequences of whitespace (spaces, newlines, tabs) were replaced with a single space, and leading/trailing whitespace was removed using `.strip()`. This created clean, single-line strings.

4. Keyword Formatting (clean_keywords function):

For the fine-tuning task, the comma-delimited keywords string was cleaned by replacing commas with spaces to form a coherent target sequence for the generative model.

5. Data Splitting:

- **For Baseline Evaluation (YAKE, KeyBERT):** A random sample of 50 records was used as a consistent test set to allow for rapid and reproducible evaluation.
- **For Fine-Tuning (T5 Model):** The entire dataset was used. It was split into a training set (80%) and a validation/test set (20%) using scikit-learn's `train_test_split`. This ensures the model is evaluated on data it has never seen during training.

4. Exploratory Data Analysis (EDA)

EDA was performed on the preprocessed training data to gain insights into the dataset's characteristics.

- **Transcription Length:** The distribution of transcription lengths is a long-tailed distribution, with most reports being between 200 and 1500 words. This highlights the need for models that can handle long contexts.
- **Keyword Analysis:** A word cloud of the keywords column revealed that terms like "surgery," "disease," "catheter," and "pain" were highly frequent, indicating common themes across medical specialties.
- **Specialty Distribution:** The dataset contains a wide variety of medical specialties, with "Surgery," "Consult" and "Cardiovascular / Pulmonary" being among the most common.

5. Methodology & Algorithms

The project followed a three-tiered approach to progressively build a more sophisticated solution, directly addressing the assignment's requirement to compare baseline models against a fine-tuned one.

5.1. Baseline 1: YAKE (Statistical Approach)

- **Algorithm:** YAKE (Yet Another Keyword Extractor) is an unsupervised statistical algorithm that scores candidate phrases based on features like word frequency, position, and co-occurrence. It requires no AI model or training.
- **Reasoning for Choice:** YAKE was chosen to establish a strong, non-AI baseline. It answers the question: "How well can we do with a fast, purely statistical method?" This provides a crucial benchmark to justify the use of more complex models.

5.2. Baseline 2: KeyBERT (Pre-trained Language Model Approach)

- **Algorithm:** KeyBERT utilizes a pre-trained SentenceTransformer model to find keywords. It works by converting the entire transcription and candidate phrases into vector embeddings and then selecting the candidates whose meaning is most similar to the overall document's meaning.
- **Reasoning for Choice:** This fulfills the requirement to **incorporate a language model externally**. We used `pritamdeka/S-BioBert-snli-multinli-stsb`, a model pre-trained on biomedical text, to ensure strong domain-specific semantic understanding. KeyBERT represents a powerful, "off-the-shelf" AI baseline to compare against our custom-trained solution.

5.3. Advanced Solution: Fine-Tuning a T5 Model (Generative Approach)

- **Algorithm:** This approach reframes keyword extraction as a **sequence-to-sequence** task. We used the t5-base model, a powerful Text-to-Text Transfer Transformer from Google. The model is trained to read an input sequence (the transcription) and generate an output sequence (the keywords).
- **Reasoning for Choice:** This was the core of the project, directly fulfilling the **Train/Fine-tune on given domain-specific dataset** requirement. The hypothesis was that by training the model on our specific data, it would learn not only the relevant medical concepts but also the specific style and structure of the desired keyword output. This represents the most advanced and customized solution.

6. Challenges, Findings, and Results

6.1. Challenge: Baseline Model Performance

A significant early finding was that the sophisticated KeyBERT model performed at a very similar level to the much simpler YAKE model. This presented a challenge, as it indicated that simply applying a powerful, general-purpose AI model was not a guaranteed path to superior performance. This finding highlighted that the dataset's keywords often had a strong lexical overlap with the source text, a characteristic that favored YAKE's statistical nature.

6.2. Solution: The Power of Fine-Tuning

The solution to overcoming the performance ceiling of the baseline models was to implement the fine-tuning strategy. By training the t5-base model on the full dataset, we allowed it to adapt its internal parameters to the specific nuances of our data.

6.3. Final Results

The fine-tuned model demonstrated a dramatic and decisive improvement over both baselines.

Model	ROUGE-1 (F1)	ROUGE-L (F1)	Semantic Similarity
YAKE (Baseline 1)	~0.30	~0.23	~0.60
KeyBERT (Baseline 2)	~0.27	~0.20	~0.60
Fine-Tuned T5-Base	0.6223	0.5244	~0.82

Conclusion: The results unequivocally prove the effectiveness of fine-tuning for domain-specific NLP tasks. The fine-tuned T5 model's final **Semantic Similarity score of 81.8%** showcases its superior ability to understand the medical context and generate keywords that are not only lexically accurate but also semantically aligned with the ground truth. The model successfully learned the mapping from complex medical reports to concise, relevant keyword summaries, significantly outperforming the pre-trained and statistical methods.